

Astrodynamics I: Midterm Exam

April 17, 2017 5:05-6:35PM (90 minutes)

Closed Books/Notes

You may use the following formulae without proof if necessary.

$$\bullet \quad \vec{e} = \frac{1}{\mu} \left[\left(v^2 - \frac{\mu}{r} \right) \vec{r} - (\vec{r} \cdot \vec{v}) \vec{v} \right], \quad e = \sqrt{1 + \frac{2Eh^2}{\mu^2}}$$

1. (30) Two-Body Problem

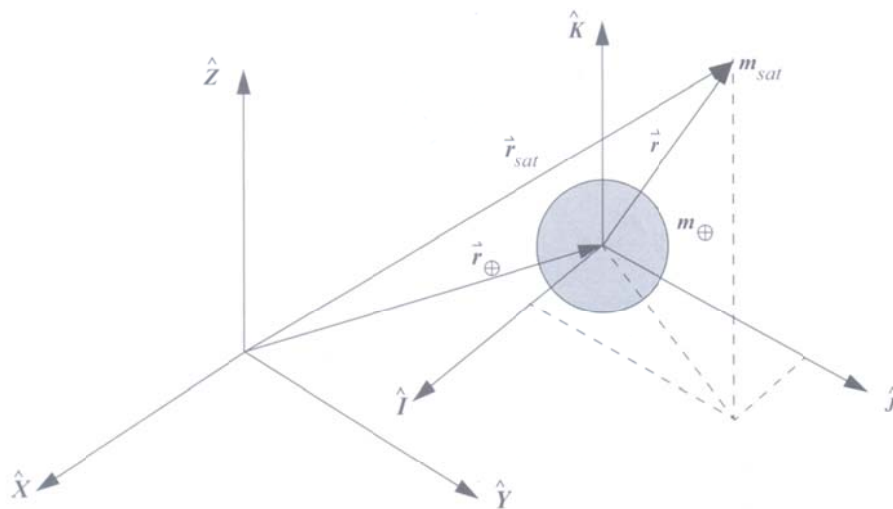


Figure 1-12. Geometry for Two Bodies in an Inertial Reference Frame. XYZ is assumed to be an inertial coordinate system. IJK is displaced from XYZ, but does not rotate or accelerate with respect to XYZ.

- A. (5) State precisely four underlying assumptions for deriving the two-body equations of motion.
 - B. (10) Using the notations in Figure 1-12, derive the equations of motion of a satellite (sat) with respect to the primary body ($\oplus$) in the 2 body problem.
 - C. (5) Show that the specific angular momentum is constant.
 - D. (10) Show that the specific energy is constant.
2. (20) A spacecraft is orbiting a planet in a circular orbit with a radius of 4 Planet Radii ($a=4, e=0$). A maneuver is performed that rotates the velocity vector in the orbital plane by an angle of 45° towards the planet.
 - A. (8) Find the new semimajor axis a_n and the new eccentricity e_n following this maneuver.
 - B. (6) At what point in its new orbit is the spacecraft in terms of true anomaly (ν)?
 - C. (6) Will the spacecraft impact the planet? Justify your answer.

3. (30) f and g functions

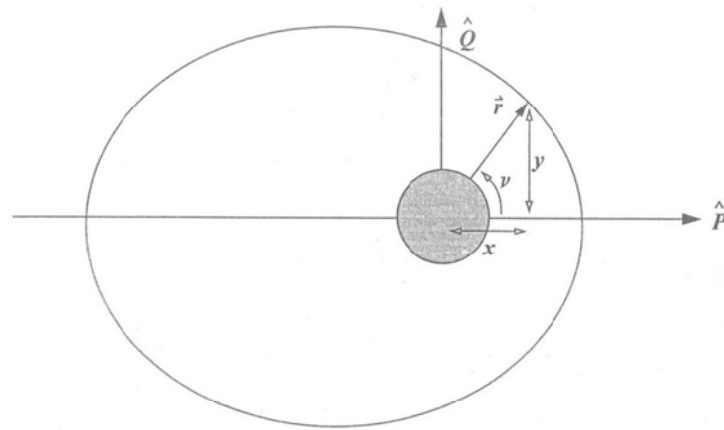


Figure 2-14. Geometry for the Perifocal Coordinate System. We can break a satellite's position into two components (x, y) in the orbital plane.

The position and velocity vectors can be expressed in perifocal coordinate system as

$$\begin{cases} \vec{r} = x\hat{P} + y\hat{Q} \\ \vec{v} = \dot{x}\hat{P} + \dot{y}\hat{Q} \end{cases}$$

It is assumed that the initial position and velocity vectors $(\vec{r}_0, \vec{v}_0)$ are known. Then, the position and velocity vectors at an arbitrary time can be also expressed as

$$\begin{cases} \vec{r} = f\vec{r}_0 + g\vec{v}_0 \\ \vec{v} = \dot{f}\vec{r}_0 + \dot{g}\vec{v}_0 \end{cases}$$

- A. (10) Derive $(f, \dot{f}, g, \dot{g})$ as functions of (x, y, x_0, y_0) and their derivatives.
 - B. (5) Discuss how to check the accuracy of (f, g) functions.
 - C. (15) Suppose that the true anomaly ν is known. Derive (f, g) as functions of the change in true anomaly $\Delta\nu = \nu - \nu_0$.
4. (20) Determine the orbital elements for an Earth orbit which has the following position and velocity vectors and sketch the orbit (set the gravitational parameter $\mu = 1$):

$$\begin{cases} \vec{r} = 1K \text{ DU} \\ \vec{v} = 1I \text{ DU/TU} \end{cases}$$